# SOROUSH AZARI

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## SUMMARY

Data analyst with an MSc in Advanced Computer Science (University of Exeter, Merit) and a background in machine learning, statistics and mathematics. I take large datasets from several sources through cleaning, quality assurance and analysis to accurate Power BI dashboards and clear written reports for non-technical readers. My most recent project uses Environment Agency rainfall and river-flow records with South West Water storm-overflow monitoring data. Founder of TiTil Studio, where I build and publish mobile apps.

## TECHNICAL SKILLS

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| **Reporting and BI** | Power BI (data modelling, DAX measures, dashboards), Excel |
| **Data analysis** | SQL (joins, window functions, CTEs), Python (pandas, scikit-learn), regression and forecasting |
| **Data quality** | Data cleaning, quality assurance checks, trend and anomaly analysis, KPI definition |
| **Communication** | Writing up findings and limitations for non-technical readers |
| **Tools** | Git and GitHub, Environment Agency Hydrology API, open government data, Supabase |

## DATA PROJECTS

**Rainfall, River Flow and Storm Overflows in Devon and Cornwall | Python, SQL, Power BI | azed.uk/water.html**

- Combined two external sources: 2015–2025 daily rainfall (7 gauges) and river flow from the Environment Agency Hydrology API, and South West Water's 2025 storm-overflow data (209,711 events at 1,179 sites).
- Ran quality assurance: resolved duplicate readings, flagged the 10.7% of readings not rated Good, and excluded 356 implausibly long overflow events, documenting each decision.
- Analysed trends, patterns and anomalies: the last 14 days of rain explain River Exe flow far better than same-day rain (correlation 0.77 against 0.34); the worst 10% of overflow sites account for about 47% of discharge hours; low-flow days cluster in 2022, 2018 and 2025.
- Built a next-day flow forecast that beats a persistence baseline on held-out data (Nash-Sutcliffe efficiency 0.85 against 0.78), a six-page Power BI dashboard with DAX measures, 11 SQL queries and a written project report.

**Credit Risk Analytics and Policy Scenario Analysis | SQL, Python, Power BI | azed.uk/credit-risk.html**

- Built a customer-level dataset by joining several relational tables and engineered affordability and behavioural features for risk segmentation.
- Analysed portfolio and segment risk, finding default risk concentrated in particular customer groups, and modelled approval-versus-risk trade-offs under different credit thresholds in an interactive dashboard.

**UK Railway Performance Analysis | Power BI, SQL, Python | azed.uk/railway.html**

- End-to-end analysis of rail performance and ticketing data, with dashboards tracking delays, pricing and route performance.

## EDUCATION

**MSc in Advanced Computer Science**
University of Exeter | Exeter, UK (Awarded with Merit) Sep 2023 – Nov 2024

- Coursework: Machine Learning, Learning from Data, Social Networks and Text Analysis, Computer Vision, High-Performance Computing, Research Methodology.
- Dissertation: Balancing Accuracy and Fairness in Logistic Regression Through Multi-Objective Optimization.

**MSc in Artificial Intelligence**
Islamic Azad University, Lahijan, Iran Nov 2015 – Jan 2018

- Modules: Machine Learning, Pattern Recognition, Neural Networks, Advanced Data Mining.
- Thesis: Human activity recognition in 3D video using deep belief networks.

**BSc in Software Engineering**
Islamic Azad University, Lahijan, Iran Sep 2010 – Oct 2015

- Included probability and statistics, mathematics and databases.

## SOFTWARE PROJECTS

- TiTil Studio (founder, 2026–present): an independent studio that designs, builds and ships mobile apps, with no ads and privacy as a default. Stack: Flutter/Dart, Supabase, Firebase.
- Holddd (live on iOS and Android): a "pause before you buy" finance app. A 24-hour timer, savings tracking, budget goals and spending insights help people turn impulse purchases into real savings. Built with Flutter and Supabase.

**Links:** titilstudio.com Holddd on the App Store

## LANGUAGES

- English: MSc taught and examined in English at the University of Exeter (2023–2024).
- Persian: native.

## REFERENCE

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